

Subreddit Recommender: Neural Collaborative Filtering (NCF) with Semantic Text Embeddings

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Abstract

We study subreddit recommendation by extending Neural Collaborative Filtering (NCF) with semantic text embeddings derived from large language models. Traditional ID-based recommender systems perform poorly under sparse interaction histories and cold-start scenarios. To address this limitation, we incorporate user- and subreddit-level textual content using transformer-based embeddings generated by Qwen3-Embedding-8B. We evaluate our models on two versions of the Engage Corpus: a single-label benchmark dataset and a more realistic multi-label extension with stricter activity constraints. Our results demonstrate that, although subreddit-level semantic text captures valuable complementary information, its inclusion does not improve the predictive performance of the collaborative filtering model in the multi-label setting, likely because the baseline model is already highly predictive.

1 Introduction

Recommender systems are essential to any modern web platform; however, standard click-based ID systems can fail for users without sufficient interaction histories. The *cold-start problem*, particularly found in new users, requires introducing additional data to uncover more detail into interaction preferences. Thus, our project introduces an auxiliary data modality of user text content for more reliable subreddit recommendation.

1.1 Task Description

- **User History:** The system identifies active engagement in several of the most popular subreddit communities such as `r/Python`, `r/MachineLearning`, and `r/DataScience`.
- **User Text:** It simultaneously extracts semantic signals from comments, detecting terms like "gradient" or "neural".

- **Goal:** The system fuses these modalities to recommend relevant unvisited communities, such as `r/NLP`.

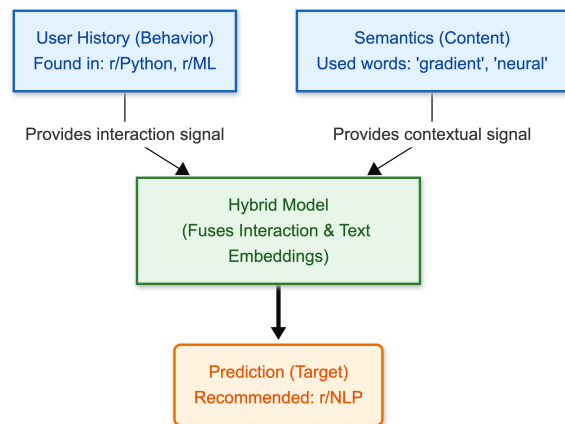


Figure 1: System workflow. The model synthesizes behavioral history and semantic content to predict relevant unvisited communities.

1.2 Formal Definition

The formal definition of the subreddit recommendation task is a top- k ranking problem. We define $\mathcal{U}$ and $\mathcal{S}$ as the sets of users and candidate subreddits, respectively.

We denote the interaction history as a sparse binary matrix $\mathbf{Y} \in \{0, 1\}^{|\mathcal{U}| \times |\mathcal{S}|}$, where $y_{us} = 1$ indicates an observed interaction. To account for sparsity concerns, we generate dense Qwen-3 semantic embeddings for user and subreddit text:

- User text embeddings $\mathbf{E}_U = \{\mathbf{e}_u \mid u \in \mathcal{U}\}$
- Subreddit text embeddings $\mathbf{E}_S = \{\mathbf{e}_s \mid s \in \mathcal{S}\}$

Our model predicts the probability of interaction by learning a scoring function $\hat{y}_{us} = f(u, s, \mathbf{e}_u, \mathbf{e}_s | \Theta)$. We optimize parameters Θ using pointwise Binary Cross-Entropy (BCE) loss

with negative sampling, to ensure that relevant unobserved subreddits are ranked higher than non-relevant ones.

1.3 Motivation

Our interest in this domain is driven by three primary factors:

- **Practical Relevance:** Recommender and ranking systems make up the backbone of content discovery on modern digital platforms. Building robust models that can handle sparse, noisy user data and cold-start interactions is a crucial skill for real-world applications.
- **Improvement Potential:** We hypothesized that recent advancements in Large Language Models (LLMs) offered a distinct opportunity to improve upon the original paper’s contextual modeling approach. Specifically, we identified that modern embedding-based methods could significantly enhance capabilities of the original architecture, capturing richer semantic understanding of user preferences.
- **Learning Opportunity:** This project serves as a comprehensive unification of our coursework in NLP in a cohesive, applied research workflow.

2 Literature Review

Recommender systems have traditionally relied on interaction-based collaborative filtering methods, such as matrix factorization, which model user-item preferences using latent ID embeddings. While effective in dense settings, these approaches struggle in sparse and cold-start scenarios where limited interaction history is available. Recent research has explored incorporating textual and contextual information to address these limitations.

Text-Aware Recommendation. Zheng et al proposed DeepCoNN, a hybrid recommender system that jointly models users and items using review text. DeepCoNN encodes aggregated user and item reviews through parallel convolutional neural networks (CNNs), whose outputs are combined using a Factorization Machine to capture

feature interactions. Their results on Yelp, Amazon, and Beer review datasets demonstrate significant improvements over traditional latent factor models, particularly in sparse settings where textual information compensates for missing interactions. This work motivates our use of aggregated user- and subreddit-level text to enrich representations beyond interaction IDs alone.

Sequential and Contextual Modeling. Sun et al introduced BERT4Rec, a Transformer-based sequential recommender that models user behavior using bidirectional self-attention. By adapting the Cloze objective from natural language processing, BERT4Rec captures both past and future contextual dependencies within user interaction sequences. The model consistently outperforms Neural Collaborative Filtering and recurrent baselines on multiple benchmarks, highlighting the effectiveness of Transformer-derived contextual representations. While our approach does not train a sequential Transformer end-to-end, it similarly leverages large pretrained language models to encode contextual information from user-generated text.

Community Representation Learning. Martin proposed community2vec, which learns vector representations of online communities based on shared user participation patterns. By constructing co-occurrence matrices over subreddit activity and applying embedding techniques such as PPMI, PCA, and GloVe, the study shows that communities occupy a meaningful semantic space. These embeddings cluster thematically related subreddits and support analogy-style reasoning, validating the premise that user participation encodes semantic relationships between communities. This work provides conceptual grounding for representing subreddits as dense vectors and motivates embedding-based similarity in recommendation tasks.

Relation to Our Work. Taken together, these studies demonstrate that recommendation performance improves when user and item representations incorporate semantic and contextual information rather than relying solely on interaction IDs. Building on this insight, our approach combines Neural Collaborative Filtering with semantic text embeddings derived from a large pretrained Transformer model. Unlike prior work that trains text encoders end-to-end, we employ general-purpose embeddings to encode aggregated

user and subreddit text histories, integrating them into a scalable recommendation pipeline. This design allows us to evaluate the extent to which modern language model embeddings can complement collaborative filtering, particularly under sparse and multi-label engagement settings.

3 Experimental Design

3.1 Data

We utilize the **Engage Corpus** [1] dataset, containing a large dataset of Reddit user and item interactions. We processed the raw corpus into two datasets with distinct preprocessing logic and constraints: **v3.5 (Single-Label)** and **v4 (Multi-Label)**. The data versions are documented in Table 1.

The raw corpus shows a heavy engagement skew. Comments comprise 90.1% of all measured activity compared to only 9.9% for posts. Despite this, we treat both as equivalent interaction signals. User activity exhibits a long-tail distribution: 61.6% of users have fewer than 10 total interactions, while only 3.8% of users have more than 500 recorded actions. This motivates our filtering choices in the v4 dataset, as users with sparse history provide an insufficient signal for generating meaningful embeddings.

Both splits utilized identical tokenization procedures. This entailed removing non-alphanumeric characters, lower-casing all text, and splitting on white space. After preprocessing, feature selection was performed using TF-IDF, selecting the top 50 discriminative words per user and subreddit to prioritize terms with high information value within the corpus, while reducing noise.

3.1.1 Dataset v3.5 (Single-Label)

This dataset was created to mirror the evaluation setup found in the **Engage Corpus** paper. We filtered the corpus to the top 5,000 subreddits and randomly sampled 6% of eligible users (who interacted with the top subreddits). We employed a temporal train-test split where the model was trained for the period Jan-Sept 2019 and evaluated for Oct-Dec 2019. The test set uses **single-label** evaluation where there is one ground-truth subreddit interaction for each user.

3.1.2 Dataset v4 (Multi-Label)

We created the v4 dataset to extend our embeddings to a more realistic setting with stricter con-

Table 1: Comparison of Dataset Preprocessing and Splits.

Feature	v3.5 (Single-Label)	v4 (Multi-Label)
Subreddit Filter	Top 5,000 subreddits	Top 2,000 subreddits
of Users	34,000	180,000
User Selection	Random 6% sample of eligible users	Strict filtering: 50-5000 posts/comments in train period
Activity Constraints	None	Must be active in both Jan-June AND July-Sept 2019
Text Processing	TF-IDF (Top 50 words per user/subreddit)	TF-IDF (Top 50 words per user/subreddit)
Training Split	Jan-Sept 2019	Jan-June 2019 (Input) July-Sept 2019 (Target)
Evaluation Split	Oct-Dec 2019	Dev: Oct & Dec 2019 Test: Nov 2019
Target Type	Single-label (1 item per user)	Multi-label (All positive interactions)

straints. We found our earlier settings in v3.5 to be overly broad in associating users to relevant subreddits upon evaluation. In v4, we applied a stricter filtering to the top 2,000 subreddits with a user activity threshold between 50 and 5,000 interactions (posts/comments) during the training period. The purpose of the threshold is to ensure every user has significant textual history to generate meaningful embeddings while removing potential bot activity (unrealistically high interaction).

Conversely to v3.5, this dataset treats recommendation as **multi-label** classification. Activity from Jan-June 2019 is used as training inputs, while the prediction targets are derived from July-Sept 2019. Activity from Oct and Dec 2019 are treated as development set. We evaluated on data from Nov 2019 to better capture diverse user preferences across multiple subreddit communities.

3.2 Evaluation Metrics

We evaluate our models using two standard ranking metrics:

Hit Rate at 10 (HR@10) Measures the fraction of users where at least one relevant (correct) subreddit appears in the top 10 predictions.

$$HR@10 = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[r_i \leq 10] \quad (1)$$

where n is the number of users, r_i is the rank of the correct subreddit for user i in the model’s ranked list, and $\mathbb{I}[\cdot]$ is the indicator function. If

the correct subreddit does not appear in the top-10 recommendations, this is treated as $r_i > 10$, where that user contributes 0 (miss) to HR@10.

NDCG at 10 (NDCG@10) Normalized Discounted Cumulative Gain accounts for the position of the hit, rewarding higher ranks in the top-10 recommendations.

$$NDCG@10 = \frac{1}{n} \sum_{i=1}^n \frac{1}{\log_2(r_i + 1)} \quad (2)$$

where r_i is the 1-indexed rank of the correct subreddit among the top-10 recommendations for user i . If the correct subreddit does not appear in the top-10 recommendations for user i , we set $r_i = \infty$, yielding a contribution of 0 for that user to NDCG@10.

Multi-Label Evaluation In the original **Engage Corpus** [1] paper, their evaluation process is single-label, meaning each user had exactly one correct item in the test period. With multi-label evaluation, users can have multiple relevant subreddits (Avg ≈ 2.38 per user) to consider multiple positive matches when scoring ranking quality. With more opportunities for a hit, these metrics are expected to increase in a multi-label setting.

3.3 Simple Baseline

As a simple, non-collaborative baseline, we implement a text-only recommender using cosine similarity over semantic embeddings. This baseline serves to isolate the predictive power of textual content alone, without leveraging user-subreddit interaction histories or learned latent factors.

User representations are constructed by aggregating the textual content of each user’s posts and comments during the training period. Subreddit representations are similarly formed by aggregating text associated with each subreddit. Both user and subreddit texts are embedded using the Qwen3-Embedding-8B model, producing dense semantic vectors. To improve efficiency and compatibility with downstream models, embeddings are projected to a reduced dimensionality.

For each user, candidate subreddits are ranked by cosine similarity between the user embedding and each subreddit embedding. This ranking directly produces a recommendation list without any learned parameters or training procedure. Evaluation is performed using the same HR@10 and NDCG@10 metrics as our neural models, allowing for direct comparison.

This baseline provides a strong semantic reference point and helps contextualize the gains achieved by incorporating collaborative filtering and neural interaction modeling. The baseline results can be found in Table 2, where we see an increase in performance to both datasets by using higher-dimensional embeddings.

Table 2: Performance of the Cosine Similarity baseline in the V3.5 and V4 datasets.

Model	v3.5 (Single-Label)		v4 (Multi-Label)	
	HR@10	NDCG@10	HR@10	NDCG@10
Cosine Sim (128-dim)	27.33	23.36	28.65	10.52
Cosine Sim (4096-dim)	32.66	28.23	35.47	13.98

4 Experimental Results

We implemented a Neural Collaborative Filtering (NCF) backbone that utilizes Generalized Matrix Factorization (GMF) and a Multi-Layer Perceptron (MLP). A representation of the model architecture can be found in figure 2 in the Appendix.

4.1 Published Baseline Implementation

We began by replicating the results of the original paper to establish a valid performance baseline. As shown in Table 3, our replication of the NCF model (without context) achieved an HR@10 of 26.12%, significantly outperforming the naive trigram logistic baseline. However, we were unable to achieve the same performance of the original paper, likely due to divergence in data preprocessing configurations.

Table 3: Baseline Replication vs. Original Paper

Model	HR@10 (%)	NDCG@10 (%)
Trigram logistic baseline	1.75	0.72
NCF (no context)	26.12	16.46
NCF + text-based context	26.33	16.29
Original Paper NCF	53.50	33.50
Original Paper NCF + Text	60.30	38.50

4.2 V3.5 Dataset (Single-Label) Performance

In the V3.5 single-label classification task, we observed that pure Cosine Similarity on high-dimensional embeddings actually outperformed the NCF + Text architecture (Model A). This suggested that for single-label tasks, the complexity of the NCF model might have been detrimental compared to direct similarity measures.

Table 4: V3.5 Single-Label Results: Cosine vs. NCF

Model	HR@10 (%)	NDCG@10 (%)
Cosine Similarity (128-dim)	27.33	23.36
Cosine Similarity (4096-dim)	32.66	28.23
NCF + Text (Model A)	15.57	09.93

4.3 V4 Dataset (Multi-Label) & Ablation Study

For the V4 multi-label dataset, we evaluated standard cosine similarity baselines alongside our NCF Neural architecture and its ablations. Table 5 presents the comprehensive results.

While the NCF Baseline (Model B) achieved the highest metrics (HR@10: 53.21%), the ablation study reveals that feature engineering techniques like Gating (Model E, G) and Chi-Square selection (Model F) generally reduced performance compared to the raw collaborative signal. However, Gating was shown to recover some performance when applied to the reduced feature set (Model G vs Model F).

Table 5: V4 Multi-Label Results: Baselines & Ablations

Model ID	Configuration	HR@10 (%)	NDCG@10 (%)
<i>Similarity Baselines</i>			
–	Cosine Similarity (128-dim)	28.65	10.52
–	Cosine Similarity (4096-dim)	35.47	13.98
<i>Neural Collaborative Filtering (NCF) Variants</i>			
Model B	NCF Baseline (No Text)	53.21	22.38
Model C	NCF + Text (TF-IDF)	52.46	22.21
Model E	NCF + TF-IDF + Gated	48.11	18.71
Model F	Chi2 Feature Selection	44.22	16.67
Model G	Chi2 + Gated	47.88	18.57

4.4 Extensions

4.4.1 Extension 1: TF-IDF Filtering

A key challenge in incorporating user-generated Reddit text into recommendation models is the high level of noise and redundancy present in raw social media content. Frequent conversational terms and domain-specific tokens can obscure discriminative signals when used directly as model inputs. To address this issue, we introduce TF-IDF (Term Frequency–Inverse Document Frequency) filtering as a lightweight textual preprocessing extension.

For each user and subreddit, we aggregate all associated posts and comments observed during training into a single document. Standard preprocessing such as tokenization, case normalization is applied. TF-IDF statistics are then com-

puted over the training corpus, and for each entity we retain only the top 50 tokens ranked by TF-IDF score. This selection strategy reduces input dimensionality while emphasizing informative keywords (e.g., *gradient*, *tensor*, *backprop*) over high-frequency but low-information terms.

The resulting filtered text representations are used both for downstream semantic embedding generation and as auxiliary textual features in Neural Collaborative Filtering (NCF) variants, without modifying the underlying model architecture or training objective. Empirically, TF-IDF filtering improves semantic focus but does not consistently improve recommendation performance, particularly in the multi-label v4 setting where strong collaborative signals already dominate, as shown in table 5. These findings motivate the exploration of richer semantic representations, which we address in subsequent extensions.

4.4.2 Extension 2: QWEN Text Embeddings

While TF-IDF filtering improves lexical focus, it remains limited to surface-level word statistics and cannot capture semantic similarity between related concepts expressed using different vocabulary. To address this limitation, we extend our approach by replacing sparse lexical representations with dense semantic embeddings derived from a large pretrained language model.

We employ the Qwen3-Embedding-8B model to generate fixed-length vector representations for both users and subreddits. For each entity, we aggregate all TF-IDF-filtered posts and comments observed during the training period into a single document and compute its embedding using the frozen Qwen encoder. We evaluate two embedding dimensionalities, 128 and 4096, to study the trade-off between representational capacity and computational efficiency. All embeddings are generated offline and remain fixed during downstream model training, ensuring scalability and preventing information leakage from evaluation periods.

These embeddings are used in two complementary ways. First, they form the basis of a text-only baseline that ranks candidate subreddits using cosine similarity between user and subreddit embeddings. Second, the embeddings are incorporated as auxiliary features in NCF variants, where they augment ID-based latent factors rather than replacing them. This design allows us to isolate the predictive value of semantic text information both independently and in conjunction with collabora-

tive signals.

Empirically, the high-dimensional Qwen embeddings ($d=4096$) achieved a notable performance peak in the single-label v3.5 setting, outperforming the neural architecture as seen in table 4. However, in the multi-label v4 setting, the integration of these embeddings into the NCF model yielded diminishing marginal returns, as shown in table 5. These results indicate that pretrained semantic embeddings are most effective in sparse or cold-start regimes, while their marginal benefit diminishes as collaborative signals become stronger.

4.4.3 Extension 3: Chi-Square Feature Selection

While TF-IDF filtering is effective at identifying rare terms, it is an unsupervised method that does take into account the predictive relationship between tokens and class labels (subreddits). Therefore, we implemented Chi-Square (χ^2) statistical filtering as an alternative feature selection method. The motivation was to use a supervised filtering method to retain tokens that possessed a strong statistical dependence with specific subreddit labels to generate higher-signal embeddings.

The Chi-Square scoring mechanism states that for a given token t and subreddit class s , the χ^2 statistic measures the difference between the observed and expected co-occurrence frequencies:

$$\chi^2(s, c) = \frac{N(AD - BC)^2}{(A + C)(B + D)(A + B)(C + D)} \quad (3)$$

where N is the total number of users, A is the frequency of token t . Other contingency table frequencies such as term not in class or term in other class are represented by B , C , and D . For each user’s history, we aggregated their scores, keeping only their top-50 terms with the highest significance.

However, this approach proved to degrade performance as shown in Table 5. Model F (Chi-Sq without gating) performed worst across all variants, with an HR@10 of 44.22%. While Chi-Square feature selection removed non-discriminative terms such as "help" or "good", it proved to be over-aggressive in filtering nuanced text signals, leading to overly generic user embeddings. Additionally, this method suffers from tunnel vision, optimized to find words exclusive to a specific class (correlation), failing to derive semantic cues for the NLP to connect closely related subreddits. A flowchart illustrating this feature selection process can be found in Figure 3 in the Appendix

4.4.4 Extension 4: Gated Fusion Architecture

We introduced a learnable Gated Fusion architecture to our model to dynamically tune the trade-off between collaborative signals and text embeddings. As we saw no improvement from incorporating equally-weighted textual features in Model C from the baseline, Gated Fusion allows the network to learn when to utilize strong text embeddings and when to ignore weak or noisy ones.

To control the information flow, the gate uses a Sigmoid activation function in the last layer of the network. Let $\mathbf{h}_{ID}$ represent the factor from the collaborative filtering branch and $\mathbf{h}_{text}$ represent the auxiliary text embedding. The fused branch $\mathbf{h}_{fused}$ is computed as:

$$\alpha = \sigma(\mathbf{W}_g[\mathbf{h}_{ID}; \mathbf{h}_{text}] + b_g) \quad (4)$$

$$\mathbf{h}_{fused} = (1 - \alpha) \odot \mathbf{h}_{ID} + \alpha \odot \mathbf{h}_{text} \quad (5)$$

where $\mathbf{W}_g$ and b_g are learnable parameters and σ is the Sigmoid function ensuring $\alpha \in [0, 1]$. When the text embedding is irrelevant, the model learns to output $\alpha \approx 0$, defaulting back to solely rely on the ID-based collaborative signal. Conversely, the model learns to output $\alpha \approx 1$ when the text provides high-fidelity, valuable information, allowing free flow of embedding signals.

The results from Table 5 demonstrate the gate’s role to filter out noisy text embeddings. In model E, the gate acted as a bottleneck to higher-quality TF-IDF embeddings, degrading HR@10 by approximately 4% compared to simple concatenation (Model C). However, in model G, the gate improved HR@10 by 3% over the non-gated variant (model D) by effectively mitigating the contribution of noisier embeddings. These findings suggest that while Gated Fusion can recover some performance by successfully managing poor-quality embeddings, it still failed to outperform the baseline NCF model. A diagram of the gated fusion mechanism can be found in Figure 4 in the Appendix

4.5 Post Hoc Analysis

The primary experimental result—that incorporating semantic text embeddings into Neural Collaborative Filtering (NCF) does not improve performance over the collaborative baseline—appears counterintuitive, given the strong standalone performance of text-based similarity. To reconcile this discrepancy, we conducted a post hoc analysis of model predictions and learned representa-

tions. The analysis yields several consistent and complementary findings.

Finding 1: Semantic text embeddings encode statistically significant predictive signal. Figure 5 compares cosine similarities between user-subreddit embedding pairs drawn from observed (positive) interactions and randomly sampled pairs. Positive pairs exhibit a substantially higher mean cosine similarity (0.529) than random pairs (0.453). A two-sample t -test rejects the null hypothesis of equal means with $p \approx 0$ to six significant figures. This result establishes that the text embeddings capture meaningful semantic alignment between users and subreddits and are predictive in isolation.

Finding 2: Text-based similarity captures information largely orthogonal to collaborative filtering. Figure 6 presents the overlap in correctly recalled positive interactions among cosine similarity, the NCF baseline, and NCF augmented with text. The overlap between cosine similarity and the NCF-based models is limited: approximately 48.9% of the interactions recovered by top- k cosine similarity are not correctly predicted by either NCF variant. This limited intersection indicates that semantic similarity identifies a distinct subset of relevant user-subreddit pairs, supporting the conclusion that text embeddings encode complementary, rather than redundant, information relative to collaborative filtering.

Finding 3: Different models excel on systematically different subreddit segments. Table 6 stratifies subreddits by the model(s) achieving recall ≥ 0.5 , revealing clear structural differences. The NCF and NCF+Text models dominate on large, high-traffic subreddits with broad appeal (e.g., `r/AskReddit`, `r/funny`, `r/news`), a pattern consistent with their strong performance on popular-item buckets shown in Figure 8. This behavior is likely driven by interaction-frequency effects and sampling bias that favor collaborative signals in dense regions of the data. In contrast, cosine similarity performs particularly well on niche subreddits characterized by specialized and distinctive vocabulary, where semantic cues provide strong disambiguation. Notably, adding text to the NCF model degrades recall on several pornographic subreddits, plausibly due to high lexical overlap across such commu-

nities, which reduces the discriminative power of text features.

Finding 4: NCF augmented with text behaves similarly to the NCF baseline. Across aggregate metrics (Tables 3, 4, and 5), as well as subgroup analyses (Figure 8), the prediction patterns of NCF+Text closely mirror those of the NCF baseline. This similarity persists despite the demonstrated standalone effectiveness and complementarity of text embeddings, suggesting that the collaborative signal dominates the learned decision function when both modalities are present.

Finding 5: Learned projection layers attenuate the contribution of text embeddings. Figure 7 analyzes the norms of learned embedding projections within the NCF+Text model. While Qwen3-Embedding-8B produces unit-normalized embeddings, their learned projections have substantially smaller norms (approximately 0.57 for users and 0.69 for subreddits). By contrast, the GMF embeddings learned purely from interaction data span a much wider range (approximately 0.1–0.5 for users and 0.5–3.0 for subreddits). This disparity indicates that the model systematically down-weights the text modality during training. Taken together with the orthogonality observed in Finding 2, this suggests that the strong collaborative signal dominates optimization, leading the model to suppress the contribution of text features despite their independent predictive value.

5 Conclusions

In this project, we investigated whether semantic textual information can improve subreddit recommendation beyond interaction-only collaborative filtering. We built an end-to-end pipeline on the Engage Corpus and evaluated across two settings: the original single-label benchmark (v3.5) and a stricter, more realistic multi-label formulation (v4). Our approach augments Neural Collaborative Filtering (NCF) with text-derived representations, including TF-IDF filtering and dense embeddings from Qwen3-Embedding-8B.

Across experiments, we find that semantic text is highly predictive in isolation: cosine similarity over high-dimensional Qwen embeddings performs strongly, and in the v3.5 single-label setting it outperforms our NCF+Text model. However, in the v4 multi-label setting, the interaction-based NCF

baseline achieves the best overall ranking metrics, and incorporating text features yields diminishing or negative marginal returns. Ablations further support this trend: Chi-square feature selection and gated fusion generally reduce performance relative to the raw collaborative signal, with gating only partially recovering accuracy after aggressive feature reduction.

Our post hoc analyses help reconcile these results. Text embeddings encode statistically meaningful signal and retrieve a substantial set of relevant user-subreddit pairs that are not captured by NCF, particularly for niche communities with distinctive vocabulary. Nevertheless, when fused into NCF, the learned projection layers attenuate the magnitude of text features, indicating that optimization is dominated by dense collaborative signals in v4. Taken together, these findings suggest that pretrained semantic embeddings are most valuable in sparse or cold-start regimes and for long-tail subreddits, while their benefit decreases as interaction histories become sufficiently informative.

Acknowledgements

References

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A Supplementary Visuals

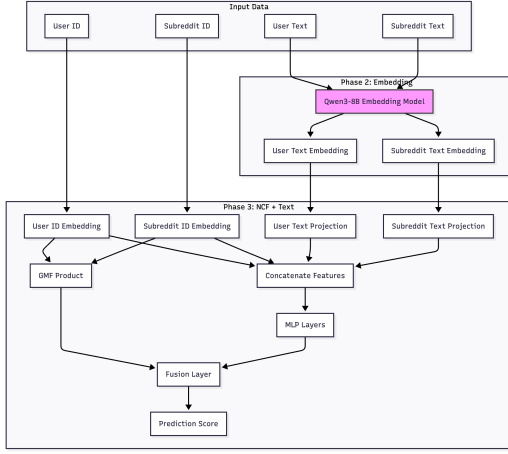


Figure 2: Visual representation of the Neural Collaborative Filtering (NCF) model architecture integrated with auxiliary text embeddings.

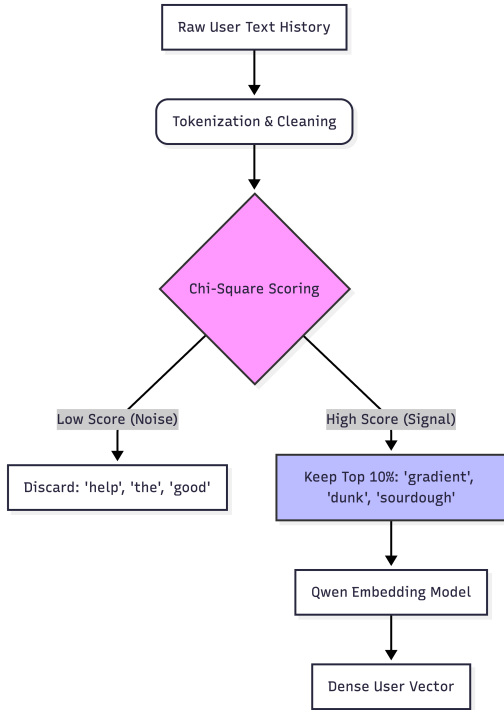


Figure 3: Flowchart illustrating the Chi-Square statistical feature selection process applied to user vocabulary.

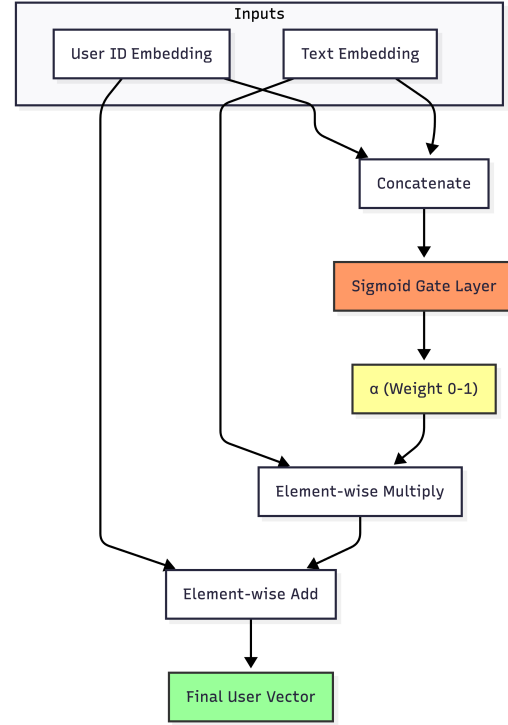


Figure 4: Diagram of the Gated Fusion mechanism used to dynamically weigh collaborative and textual features within the MLP layer.

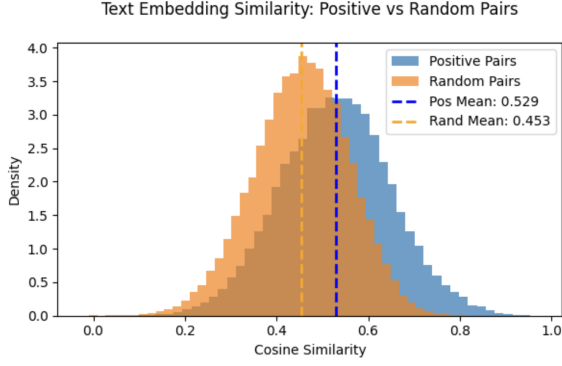


Figure 5: Model recall of positive user-subreddit interaction pairs

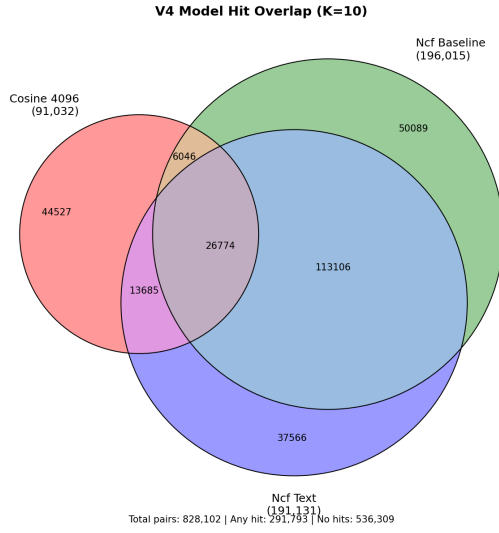


Figure 6: Model recall of positive user-subreddit interaction pairs

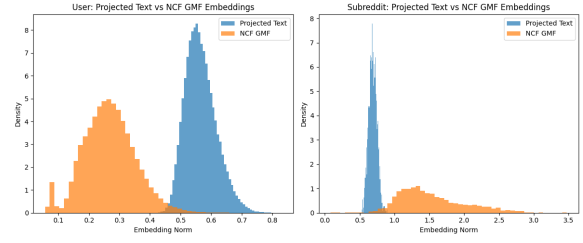


Figure 7: Learned norms from text embedding projections and collaborative filtering GMF

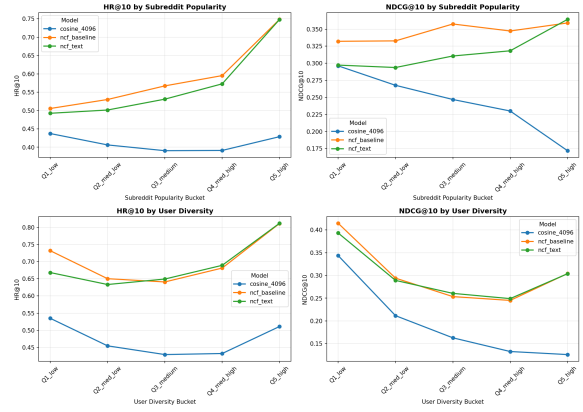


Figure 8: Model performance across data subreddit and user segments

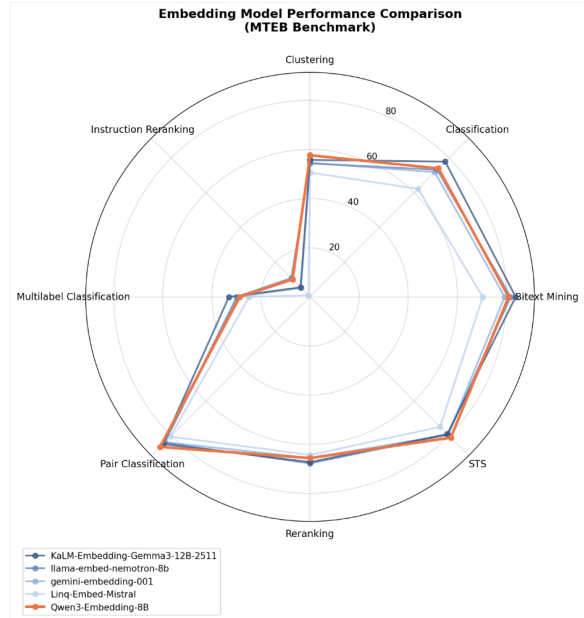


Figure 9: Qwen3-Embedding-8B performance on key embedding benchmarks

Table 6: Qualitative Analysis: Subreddits with Recall ≥ 0.5 (Venn Regions)

Venn Region	Subreddits Successfully Recalled
Only Cosine Similarity	r/pyrocynical, r/PokemonSwordAndShield, r/pokemontrades, r/summonerschool, r/RocketLeagueExchange, r/EDH, r/gundeals, r/microgrowery, r/DragonballLegends, r/Monitors, r/buildapcforme, r/HeadphoneAdvice, r/beyondthebump, r/SuggestALaptop, r/opiates, r/ar15, r/Dentistry, r/makinghiphop, r/applehelp, r/KeybaseProofs, r/fantasybball, r/guitarpedals, r/linuxquestions, r/edmproduction, r/simracing, r/UearthedArcana, r/cycling, r/churning, r/pregnant, r/btc, r/Homebrewing, r/CruciblePlaybook, r/DokkanBattleCommunity, r/buildmeapc, r/ValveIndex, r/ClashOfClansRecruit, r/ShinyPokemon, r/shortcuts, r/BDSMPersonals, r/benzodiazepines, r/ModernMagic, r/supremeclothing, r/Stims, r/Mcat, r/spikes, r/mtgfinance
Only NCF Baseline	r/todayilearned, r/Minecraft, r/Animemes, r/PetiteGoneWild, r/jobuds, r/ratemycock, r/pussy, r/MassiveCock, r/chubby, r/traps, r/BigBoobsGW, r/GaybrosGoneWild, r/EpicSeven, r/grandorder, r/asktransgender, r/Pikabu, r/FantasyPL, r/WouldYouFuckMyWife, r/Sissies, r/thebachelor, r/orangetheory, r/MtF, r/sissypersonals, r/FFBraveExvius, r/sissykik, r/transpositive, r/DadsGoneWild, r/GoneWildCD, r/ThickDick, r/gaymersgonewild
Only NCF + Text	r/aww, r/weekendgunnit
Cosine $\cap$ NCF Baseline	r/DBZDokkanBattle, r/BabyBumps, r/DesignerReps, r/bangtan, r/FashionRepsBST
Cosine $\cap$ NCF + Text	r/dirtytenpals, r/mechmarket, r/MaddenUltimateTeam, r/muacjdiscussion
NCF Baseline $\cap$ NCF + Text	r/AskReddit, r/funny, r/memes, r/pics, r/gaming, r/dankmemes, r/PewdiepieSubmissions, r/AmItheAsshole, r/worldnews, r/politics, r/news, r/teenagers, r/gonewild, r/nba, r/nfl, r/soccer, r/fantasyfootball, r/gonewild30plus, r/AsiansGoneWild, r/RealGirls, r/cock, r/penis, r/FemBoys
All Three Models	r/FIFA, r/FashionReps, r/dirtykikpals, r/Repsneakers, r/Market76, r/GaySnapchat